Check / Calc Sheet

VBF-T4R2-CALC-01 | case t4_r2

Finalist E3-porheadroom | formulas with sources, consistent with design_spec.md | 2026-08-26

All values are mechanically taken from tool output JSONs under cell/ or computed by the stated formulas. Compaction density carries the explicit /1000 kg/m3 -> g/cm3 unit conversion.

Input parameters (finalist E3, params_E3.json)

Parameter	Value	Unit	Source
Nominal cell capacity	5	Ah	parameter set Nominal cell capacity [A.h]
Measured 1C capacity 25 C (DFN)	5.03918	Ah	cell/r5_E3_1c_dfn.json (re-nominalization +0.78 %, kept)
Voltage window	2.5 - 4.2	V	parameter set Lower/ Upper voltage cut-off [V]
Electrode height	0.065	m	parameter set Electrode height [m]
Electrode width	1.58	m	parameter set Electrode width [m]
Positive thickness	7.56e-05	m	parameter set
Negative thickness	8.52e-05	m	parameter set
Separator thickness	8e-06	m	parameter set (R2: 12 um -> 8 um)
Positive CC thickness	1.2e-05	m	parameter set (R2: 16 um -> 12 um)
Negative CC thickness	8e-06	m	parameter set (R2: 12 um -> 8 um)
Stack thickness (sum)	0.0001888	m	75.6+85.2+8+12+8 um
Positive porosity	0.28	-	parameter set (R2 compression)
Negative porosity	0.28	-	params_E3.json (R5: 0.22 -> 0.28 headroom)
Separator porosity	0.47	-	parameter set
Positive particle radius	3.5e-06	m	parameter set

Parameter	Value	Unit	Source
Negative particle radius	3e-06	m	parameter set
Positive material density (derived)	3262	kg/m3	layer_kg_m2 / (t x (1-eps))
Negative material density (derived)	1657	kg/m3	layer_kg_m2 / (t x (1-eps))
Al density	2700	kg/m3	parameter set
Cu density	8960	kg/m3	parameter set
Positive active-material volume fraction	0.665	-	parameter set Positive electrode active material volume fraction
Negative active-material volume fraction	0.75	-	parameter set Negative electrode active material volume fraction
Electrolyte conductivity	1.6	S/m	params_E3.json override (transport escalation, ESTIMATE)
Electrolyte diffusivity	3.2e-10	m2/s	params_E3.json override (ESTIMATE)
Cation transference number	0.6	-	params_E3.json override (ESTIMATE)
Electrolyte initial concentration	1000	mol/m3	parameter set
Electrolyte density (fill calc)	1200	kg/m3	literature default, annotated (set has none)
Total heat transfer coefficient	80	W/(m2 K)	params_E3.json (liquid-cold-plate class)
Cell volume (mechanical refresh)	1.93898e-05	m3	R4 procedure: sum(thickness_outer surfaces) x area
Electrode area	0.1027	m2	0.065 m x 1.58 m
Cold-soak initial temperature	253.15	K	lowT protocol Initial temperature [K]

Capacity and energy

Quantity	Formula	Value	Unit	Source
1C capacity, 25 C	Q1C = integral(I)dt over 1C_discharge, DFN	5.03918	Ah	cell/r5_E3_1c_dfn.json
1C capacity, -20 C cold-soak	QlowT = integral(I)dt over lowT_discharge (T0=253.15 K), DFN	5.00782	Ah	cell/r5_E3_lowt_dfn.json
Retention	QlowT / Q1C = 5.0078 / 5.0392	0.993776	-	cell/r5_E3_retention_dfn.json (>= 0.95 required)
Discharge energy	E = integral(V*I)dt over 1C_discharge	18.1647	Wh	cell/r5_E3_energy_dfn.json
Midpoint voltage	V at 50 % discharged capacity	3.84734	V	cell/r5_E3_energy_dfn.json midpoint_voltage_v
DCR (informational)	(start OCV - V10%) / I1C	0.0020689	ohm	cell/r5_E3_energy_dfn.json dcr_ohm

Energy density vs criteria

uantity	Formula	Value	Unit	Threshold / verdict	Source
ravimetric energy ensity	18.1647 Wh / 0.0395362 kg (electrolyte excluded)	459.446	Wh/kg	>= 327.18, PASS +132.3	cell/r5_E3 energy_df .json
olumetric energy ensity	18.1647 Wh / 0.01938976 L (sum layer thickness x area)	936.822	Wh/L	>= 880, PASS +56.8	cell/r5_E3 energy_df .json

N/P ratio and mass

Quantity	Formula	Value	Unit	Source
Anode full-window capacity	c_max_neg x 0.9014 x F/3600 x t_neg x 0.75 x 0.1027	5.253	Ah	C/10 DFN stoich span + eps_active closure (scripts/calc_np_ratio.py)
Cathode full-window capacity	c_max_pos x (0.8531-0.2700) x F/3600 x t_pos x 0.665 x 0.1027	5.092	Ah	C/10 DFN stoich span + eps_active closure
N/P ratio	5.253 / 5.092	1.03162	-	positive-limited (cathode exhausts first)
Inventory closure check	electrodes transfer 5.0384 Ah (1C span); computed 5.0385 / 5.0387	5.0384	Ah	consistent at 0.1 % (diag_thermal_variant.py)
Mass - positive active	layer_kg_m2 x 0.1027	0.0182	g	calc-energy layer_kg_m2
Mass - negative active	layer_kg_m2 x 0.1027	0.0104	g	calc-energy layer_kg_m2

Quantity	Formula	Value	Unit	Source
Mass - Al CC	layer_kg_m2 x 0.1027	0.0033	g	calc-energy layer_kg_m2
Mass - Cu CC	layer_kg_m2 x 0.1027	0.0074	g	calc-energy layer_kg_m2
Mass - separator	layer_kg_m2 x 0.1027	0.0002	g	calc-energy layer_kg_m2
Mass - electrolyte fill	pore volume 5.0101e-6 m3 x 1200 kg/m3	0.006	g	pore-volume formula (literature density)
Total mass (no electrolyte) = calc-energy mass_kg	sum of layer masses	0.0395	g	matches mass_kg 0.0395362

Process design parameters

Parameter	Formula	Value	Unit	Notes
Positive areal density	layer_kg_m2 x 1000	177.557	g/m2	check: t x (1-eps) x rho = 75.6e-6 x 0.72 x 3262 = 177.6
Negative areal density	layer_kg_m2 x 1000	101.647	g/m2	check: 85.2e-6 x 0.72 x 1657 = 101.6
Positive compaction density	rho x (1-eps) / 1000 (kg/m3 -> g/cm3)	2.34864	g/cm3	unit conversion /1000 annotated
Negative compaction density	rho x (1-eps) / 1000	1.19304	g/cm3	unit conversion /1000 annotated
Electrolyte fill amount	pore volume x 1.2 g/cm3 x fill factor 1.0	0.00601214	g/cell	pore volume 5.0101e-6 m3 (eps-weighted layers x area)
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	-	-	design recommendation (production value requires tuning, annotated)